and have to be modelled as part of the simulation process. Our presentation of this topic follows that of [19] and earlier works [69, 85, 204, 276] which treat the non-Markovian Generalized Langevin equation as a Markov system in an extended phase space.

The equations of Generalized Langevin dynamics are normally written in the form

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{M}^{-1}\boldsymbol{p}\mathrm{d}t, \tag{8.57}$$

$$\mathrm{d}\boldsymbol{p} = -\nabla U(\boldsymbol{q})\mathrm{d}t - \int_0^t \boldsymbol{\Gamma}(t-s)\boldsymbol{p}(s)\mathrm{d}s\mathrm{d}t + \mathrm{d}\mathbf{B}, \tag{8.58}$$

subject to initial conditions $\boldsymbol{q}(0) = \boldsymbol{q}_0, \boldsymbol{p}(0) = \boldsymbol{p}_0$. Here $\boldsymbol{\Gamma}(t) = (\gamma_{ij}(t))$ is a $\mathrm{N_c} \times \mathrm{N_c}$ matrix-valued function of t which we refer to as the "memory kernel". The term

$$\boldsymbol{F}_\mathrm{d} = -\int_0^t \boldsymbol{\Gamma}(t-s)\boldsymbol{p}(s)\mathrm{d}s\mathrm{d}t$$

represents a drag force acting on the system due to the interaction with the particles of the bath. $\mathrm{d}\mathbf{B}$ is the increment of the colored-noise (non-Gaussian) stochastic process. Often this increment is treated formally as an impulse $\mathbf{R}(t)\mathrm{d}t$, where $\mathbf{R}$ is referred to as a "random force," which calls to mind the origin of the stochastic term as the model for interaction with a large (but finite) heat bath. We use this language for consistency with the literature. The driving forces are correlated in a way that is described by the fluctuation-dissipation theorem [202, 402]. This states that at temperature T the components of the random forces satisfy

$$\langle \mathrm{R}_i(t+s)\mathrm{R}_j(t)\rangle = \mathrm{k_B T}\,\gamma_{ij}(s)\delta_{ij}, \quad s \geq 0. \tag{8.59}$$

Here, δ_{ij} is the Kronecker delta, and $\mathrm{k_B}$ is Boltzmann's constant.

Numerical methods for treating (8.57)–(8.58) include calculation of the drag force $\boldsymbol{F}_\mathrm{d}$ which requires a convolution of the momenta with $\boldsymbol{\Gamma}(t)$ and therefore storage of some portion of the time-history of $\boldsymbol{p}$. Typically we will compute a discrete sequence of $\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots$, and so it will be necessary to store a collection of, say $\ell > 0$, previous momentum vectors. This may be somewhat demanding in a large scale simulation setting. Moreover, computing $\mathbf{R}$ requires the generation of a sequence of random numbers with prescribed correlation, as specified in Eq. (8.59). Both these issues may be addressed sensibly using the technique of Prony series and an extended variable representation

In the remainder of this discussion, for the purpose of simplified presentation, we confine ourselves to the simple case of a scalar memory kernel, writing

$$\boldsymbol{\Gamma}(t) = \gamma(t)\boldsymbol{I}.$$